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Kohl; Peter et al.

Foldable hand-held tool

Abstract

A hand-held tool (**100**) includes a handle (**104**), and a blade (**102**) pivotably coupled to the handle at a proximal end (**105**). The blade includes a plurality of apertures (**116, 118, 120, 122**). The hand-held tool further includes a locking member (**438**) is configured to detachably engage with one of the plurality of the apertures to retain the blade at an angular position relative to the handle. The locking member is provided on a bracket member coupled to the handle, the bracket member being configured to at least partially enclose the proximal end of the blade. The bracket member is spring biased such that the locking member engages with one of the plurality of the apertures to retain the blade at an angular position relative to the handle. Further, the bracket member is movable such that the locking member disengages from one of the plurality of the apertures to allow the blade to pivot relative to the handle.

Inventors:	Kohl; Peter (Illertissen, DE), Missel; Patrick (Westerstetten, DE), Reh; Bernhard (Laichingen, DE)
Applicant:	HUSQVARNA AB (Huskvarna, SE)
Family ID:	1000008541925
Assignee:	HUSQVARNA AB (Huskvarna, SE)
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Primary Examiner: Kasztejna; Matthew J

Attorney, Agent or Firm: Burr & Forman

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a foldable hand-held tool.

BACKGROUND

(2) Foldable hand-held tools, such as handsaws, are well known in the art. A foldable hand-held tool includes a handle and a blade pivotably coupled to the handle. The blade may be retained in multiple angular positions relative to the handle. Typically, a user may unlock the blade, rotate the blade to a desired angular position, and lock the blade in that position.

(3) U.S. Pat. No. 5,924,210 granted on Jul. 20, 1999 discloses a saw including a handle and a saw blade pivotably coupled to the handle and having two or more orifices. The handle includes a spring-biased projection for engaging with the orifices and for securing the saw blade at a selected angular position relative to the blade. A knob is slidably engaged in the handle for disengaging the projection from the orifices and for allowing the saw blade to rotate relative to the handle. During adjustment of the saw blade, a force required to keep the projection disengaged may be quite high causing inconvenience to a user. Further, during operation of the saw, forces acting on the saw blade may be directly transferred to the projection. This may result in premature wear of the projection.

(4) Therefore, in light of the foregoing, there is a need for an improved foldable hand-held tool.

SUMMARY

(5) In view of the above, it is an objective of the present invention to solve or at least reduce the problems discussed above. The objective is at least partially achieved according to a hand-held tool, in an embodiment of the present invention. The hand-held tool includes a handle, and a blade pivotably coupled to the handle at a proximal end. The blade includes a plurality of apertures. The hand-held tool further includes a locking member that is configured to detachably engage with one of the plurality of the apertures to retain the blade at an angular position relative to the handle. The locking member is provided on a bracket member coupled to the handle, the bracket member being configured to at least partially enclose the proximal end of the blade. The bracket member is spring

biased such that the locking member engages with one of the plurality of the apertures to retain the blade at an angular position relative to the handle. Further, the bracket member is movable such that the locking member disengages from one of the plurality of the apertures to allow the blade to pivot relative to the handle. During an operation, forces acting on the blade may be partly transmitted from the proximal end to various components of the hand-held tool in contact with the bracket member. Consequently, an area of dissipation of the forces is increased. Therefore, the forces may be dissipated without causing increased wear to the blade, the locking member, the bracket member, and/or various other components of the hand-held tool.

(6) In an embodiment, the bracket member comprises two lateral portions, each lateral portion being connected to each other at one end. Further, the two lateral portions at least partially enclose the proximal end of the blade. The two lateral portions may increase a dissipation of forces from the blade to various components.

(7) In an embodiment, the locking member is integral with the bracket member. In an alternative embodiment, the locking member is a pin coupled to one of the lateral portions. The bracket member may therefore prevent any undesired movement of the locking member.

(8) In an embodiment, the bracket member further includes at least one first coupling portion, the at least first coupling portion configured to movably engage with a bracket support of the handle. In another embodiment, the bracket member further includes a second coupling portion, the second coupling portion being configured to engage with a connecting portion of the handle. The first and the second coupling portions may substantially prevent any rotational movement of the bracket member.

(9) In an embodiment, a spring member is disposed between the bracket member and the handle. The spring member is configured to bias the bracket member such that the locking member engages with one of the plurality of the apertures.

(10) In an embodiment, an actuating member is configured to move the bracket member such that the locking member disengages from one of the plurality of the apertures. In another embodiment, the actuating member at least partly projects from an outer surface of the handle. In a further embodiment, the actuating member includes an inner boundary and a central member, the inner boundary and/or the central member being configured to contact the bracket member. The inner boundary and/or the central member may increase an area of contact between the actuating member and the bracket member. Thus, a user may have to exert lower force in order to move the bracket member via the actuating member. Further, the actuating member may be conveniently actuated by a user since the actuating member at least partly projects from an outer surface of the handle.

(11) In an embodiment, the plurality of the apertures of the blade includes at least one working position aperture corresponding to a working position of the blade, the blade extending from the handle in the working position; and a retracted position aperture corresponding to a retracted position of the blade, the blade at least partially retracted within the handle in the retracted position. Further, in another embodiment, the handle includes a slot configured to at least partially receive the blade in the retracted position.

(12) In an embodiment, the blade further comprises a non-circular aperture, the non-circular aperture being configured to engage with a corresponding non-circular portion of a screw. The engagement of the non-circular aperture of the blade with the non-circular portion of the screw may prevent any relative rotational movement between the screw and the blade.

(13) In an embodiment, the screw engages with a bushing, the screw and the bushing being rotatable relative to the handle. The screw and the bushing may facilitate a pivoting of the blade when the locking member is disengaged from one of the plurality of the apertures of the blade.

(14) In an embodiment, the handle further comprises a detachable cover member. The detachable cover member may facilitate assembly and/or maintenance of the hand-held tool.

(15) In an embodiment, the hand-held tool is one of a handsaw or a knife.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention will be described in more detail with reference to the enclosed drawings, wherein:
- (2) FIG. 1 illustrates a perspective view of a hand-held with a blade in a middle working position, according to an embodiment of the present invention;
- (3) FIG. 2 illustrates a perspective view of the hand-held tool with the blade in an upper working position, according to the embodiment of FIG. 1;
- (4) FIG. 3 illustrates a perspective view of the hand-held tool with the blade in a lower working position, according to the embodiment of FIG. 1;
- (5) FIG. 4 illustrates a side view of the hand-held tool with the blade in a retracted position, according to the embodiment of FIG. 1;
- (6) FIG. 5 illustrates an exploded view of the hand-held tool, according to an embodiment of the present invention;
- (7) FIG. 6 illustrates a sectional view of the hand-held tool along a plane A-A in FIG. 4, according to an embodiment of the present invention;
- (8) FIG. 7 illustrates a sectional view of the hand-held tool along a plane B-B in FIG. 4, according to an embodiment of the present invention;
- (9) FIG. 8 illustrates a sectional view of the hand-held tool along a plane C-C in FIG. 4 with the blade retained, according to an embodiment of the present invention; and
- (10) FIG. 9 illustrates a sectional view of the hand-held tool along the plane C-C with the blade free to pivot, according to an embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

- (11) The present invention will be described more fully hereinafter with reference to the accompanying drawings, in which example embodiments of the invention incorporating one or more aspects of the present invention are shown. This invention may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. For example, one or more aspects of the present invention can be utilized in other embodiments and even other types of structures and/or methods. In the drawings, like numbers refer to like elements.
- (12) Certain terminology is used herein for convenience only and is not to be taken as a limitation on the invention. For example, “upper”, “lower”, “front”, “rear”, “side”, “lateral”, “upwards”, “downwards”, “forward”, “backward”, “sideward”, “left”, “right”, “horizontal”, “vertical”, “upward”, “inner”, “outer”, “inward”, “outward”, “top”, “bottom”, “higher”, “above”, “below”, “central”, “middle”, “intermediate”, “between”, “end”, “adjacent”, “proximate”, “near”, “distal”, “remote”, “radial”, “circumferential”, or the like, merely describe the configuration shown in the Figures. Indeed, the components may be oriented in any direction and the terminology, therefore, should be understood as encompassing such variations unless specified otherwise.
- (13) FIGS. 1-9, illustrate various configurations of a hand-held tool **100** (hereinafter referred to as “the tool **100**”), according to an embodiment of the present invention. The tool **100** is embodied as a handsaw in FIGS. 1-9. However, in various other embodiments, the tool **100** may be a knife. The tool **100** may be used for cutting various objects, for example, but not limited to, branches, logs etc.
- (14) The tool **100** includes a blade **102** and a handle **104**. In an embodiment of the present invention, the blade **102** is pivotably coupled to the handle **104** at a proximal end **105** of the blade **102**. The blade **102** includes a cutting edge **106** with serrations. Alternately, the cutting edge **106** may be smooth (not shown). The blade **102** further includes an eyelet **107** that may facilitate grasping of the blade **102** in order to pivot the blade **102** relative to the handle **104**.

(15) The handle **104** includes a first portion **108** and a second portion **110**. The proximal end **105** of the blade **102** is coupled with the first portion **108**. The second portion **110** may be used for gripping by a user. The second portion **110** includes an end support **112** that may prevent a user's hand from slipping off the handle **104**. The first portion **108** includes a first slot **114**. The blade **102** extends from a portion of the first slot **114** disposed in the first portion **108**. Further, the second portion **110** includes a second slot **115**. In an embodiment, the first slot **114** and the second slot **115** may be continuous from the first portion **108** to the end support **112**. Further, as illustrated in FIG. 4, the blade **102** at least partly retracts into a portion of the second slot **115**. Further, the second portion **110** may include a grip (not shown) in order to facilitate grasping. In an embodiment, the handle **104** may be a single piece with a detachable cover (shown in FIG. 5) in the first portion **108**. In an alternate embodiment, the handle **104** may include two halves that are joined together.

(16) In an embodiment, the blade **102** is retained at a plurality of angular positions relative to the handle **104**. As illustrated in FIG. 1, the blade **102** is retained in a middle working position. In the middle working position, the blade **102** may extend from the handle **104** and is oriented substantially parallel relative to the first portion **108** of the handle **104**. As illustrated in FIG. 2, the blade **102** is retained in an upper working position. In the upper working position, the blade **102** may be angled upwards relative to the first portion **108** of the handle **104**. As illustrated in FIG. 3, the blade **102** is retained in a lower working position. In the lower working position, the blade **102** may be angled downwards relative to the first portion **108** of the handle **104**. As illustrated in FIG. 4, the blade **102** is retained in a retracted position. In the retracted position, the blade **102** may be at least partially retracted within the second slot **115** (shown in FIG. 1) of the handle **104**.

(17) The various angular positions of the blade **102**, as described above, are purely exemplary in nature, and the blade **102** may have any number and/or type of angular positions relative to the handle **104** within the scope of the present invention.

(18) FIG. 5 illustrates an exploded view of the tool, according to an embodiment of the present invention. The proximal end **105** of the blade **102** includes a lower working position aperture **116**, a middle working position aperture **118**, an upper working position aperture **120**, a retracted position aperture **122**, and a non-circular aperture **124** (hereinafter referred to as the apertures **116**, **118**, **120**, **122** and **124**). In an embodiment, the apertures **116**, **118**, **120** and **122** correspond to the lower working position, the middle working position, the upper working position, and the retracted position of the blade **102**, respectively. Further, the apertures **116**, **118**, **120** and **122** are substantially circular. The blade **102** pivots about the aperture **124**. The apertures **116**, **118**, **120**, **122** and **124** are purely exemplary in nature, and the blade **102** may include any other type and/or number of apertures within the scope of the present invention.

(19) The handle **104** includes a cover member **126** that is detachably coupled to the first portion **108** by a fastener **127**. The fastener **127** passes through a hole **128** in the cover member **126** and is received by a cover coupler **129** of the first portion **108**. Further, a latch **130** of the first portion **108** may detachably engage with a corresponding part (not shown) of the cover member **126**.

Alternatively, the cover member **126** may be attached to the first portion **108** by any other detachable means, such as, snap-fit couplings, or the like. The cover member **126** may enable easy assembly and also convenient access to the inner volume of the first portion **108**. In an alternative embodiment, the handle **104** may include two halves that are joined together.

(20) FIG. 6 illustrates a sectional view of the tool **100**. Referring to FIGS. 5 and 6, a ring member **202** is received by a recessed portion **204** of the handle **104**. The ring member **202** and the recessed portion **204** include aligned holes **206** and **208**, respectively. The hole **206** of the ring member **202** may be substantially non-circular, whereas the hole **208** of the recessed portion **204** may be substantially circular. A screw **209** passed through the holes **206** and **208**, and engages with the aperture **124**, and a bushing **210**. The screw **209** includes a head **211**, a first non-circular portion **212**, a second non-circular portion **214**, a first circular portion **216** and a second circular portion **218**. A width of each of the portions of the screw may progressively decrease from the head **211** to

the second circular portion **218**. The first non-circular portion **212** at partly engages with the hole **206** of the ring member **202**. The ring member **202** may provide support to the screw **209** to facilitate rotation of the blade **102**. The engagement between the first non-circular portion **212** and the hole **206** may substantially prevent any relative rotation between the ring member **202** and the screw **209**. Further, the head **211** is retained by a stepped portion **219** of the ring member **202**. The first circular portion **216** at least partly engages with the hole **208** of the recessed portion **204**. This may enable relative rotation between the screw **209** and the handle **104**. The second non-circular portion **214** of the screw **209** at least partly engages with the aperture **124** of the blade **102**. This may substantially prevent any relative rotation between the blade **102** and the screw **209**. The second non-circular portion **214** of the screw **209** also engages with a first circular portion **220** of the bushing **210**. A diameter of the first circular portion **220** of the bushing **210** may be large enough to enclose the second non-circular portion **214** of the screw **209**. Further, the second circular portion **218** of the screw **209** engages with a second circular portion **221** of the bushing **210**. In an embodiment, the first circular portion **220** of the bushing **210** may be wider than the second circular portion **221**. This may provide additional support to the screw **209** to facilitate rotation of the blade **102** relative to the handle **104**. Therefore, the ring member **202**, the screw **209** and the bushing **210** may rotate along with the blade **102** along a pivot axis P relative to the handle **104**. Further, a head **224** of the bushing **210** is retained within a hole **222** of the cover member **126**. In an embodiment, a portion of one of the screw **209**, the recessed portion **204**, and the bushing **210** may include a low friction material. Various details of the screw **209**, the ring member **202**, the recessed portion **204**, and the bushing **210**, as described above are purely exemplary in nature, and any other type of pivot connections may be envisioned between the blade **102** and the handle **104** without deviating from the scope of the present invention. In an embodiment, two blade support members **225** are provided on both sides of the first slot **114**. The blade support members **225** may provide additional support to the blade **102** so that the blade **102** may not tilt laterally.

(21) FIGS. 7-9 illustrate sectional views of the tool **100** along different planes in FIG. 4. Referring to FIGS. 5 and 7, an actuating member **302** is at least partly received within an aperture **304** of the first portion **108**. The aperture **304** may conform to a shape of at least a portion of the actuating member **302**. The actuating member **302** may be slidable into the aperture **304**. An outer portion **305** of the actuating member **302** at least partly projects from an outer surface **306** of the first portion **108**. As illustrated in FIG. 1, the outer portion **305** includes grips **307**.

(22) Referring back to FIGS. 5 and 7, the actuating member **302** includes an inner boundary **308** and a central member **310** projecting from inner surface of the actuating member **302**. In an embodiment, at least a portion of the actuating member **302** including the inner boundary **308** may be larger than the aperture **304** so that the actuating member **302** is retained within the first portion **108** of the handle **104**. Further, a bracket member **402** is disposed adjacent to the actuating member **302** within the handle **104**. The bracket member **402** includes first and second lateral portions **404**, **406**. The first and second lateral portions **404**, **406** are connected to each other at one end by a top portion **408**. The other end of each of the first and second lateral portions **404**, **406** are open. Front and rear ends of the first and second lateral portions **404**, **406** may also be open. Therefore, the first and second lateral portions **404**, **406** define a channel **410** between them. The channel **410** is delimited by the top portion **408** at one end and is open at the other end. The channel **410** may also be open at front and rear ends. As illustrated in FIG. 7, the channel **410** at least partially encloses the proximal end **105** of the blade **102**.

(23) As illustrated in FIG. 5, each of the first and second lateral portions **404**, **406** includes a horizontal bottom edge **412** and an inclined bottom edge **414** (only for the second lateral portion **406** is shown). The horizontal bottom edge **412** of at least the first lateral portion **404** may be at least partially disposed on a horizontal planar portion **416** of the first portion **108**. The horizontal planar portion **416** may be located adjacent to the recessed portion **204**. Further, an inclined planar portion **418** is connected to the horizontal planar portion **416**. Multiple low projections **419** are

provided on the inclined planar portion **418**. Each of the low projections **419** is adjacent to a high projection **420**. The inclined bottom edge **414** of the first lateral portion **404** may be at least partially disposed on the low projections **419**. An outer surface of the first lateral projection **404** may contact the high projections **420**. Further, multiple top projections **424** are disposed on the first portion **108** of the handle **104**. A part of the top portion **408** of the bracket member **402** may contact the top projections **424**. The cover member **126** may also include one or more projections (not shown) that may contact the top portion **408** of the bracket member **402**. The top projections **424** and the low projections **419** may secure the bracket member **402** between them. The bracket member **402** may be therefore not allowed to pivot. Further, as illustrated in FIG. 7, in an embodiment, the inner boundary **308** and the central member **310** of the actuating member **302** contact the first lateral projection **404**. In an alternative embodiment, only one of the inner boundary **308** and the central member **310** may contact the first lateral projection **404**.

(24) As illustrated in FIG. 5, the second lateral portion **406** includes a first coupling portion **426**. The first coupling portion **426** may include a substantially circular projection configured to movably engage with a bracket support (not shown) provided on the cover member **126** of the handle **104**. The bracket member **402** further includes a second coupling portion **428** and a third coupling portion **430**. The second coupling portion **428** and the third coupling portion **430** may also include substantially circular apertures. Further, the second coupling portion **428** includes multiple projections **431** on an outer surface. Referring to FIG. 7, in an embodiment, the circular aperture of the second coupling portion **428** is movably coupled to a connecting portion **432** of the cover member **126**. Further, a spring member **434** is at least partly disposed on the outer surface of the second coupling portion **428**, and between an inner surface **436** of the cover member **126** and an outer surface of the second lateral portion **406**. Ends of the spring member **434** may be connected to the second lateral portion **406** and the inner surface **436** of the cover member **126**. The projections **431** (illustrated in FIG. 5) may reduce contact between the spring member **434** and the outer surface of the second coupling portion **428**. This may reduce wear to the second coupling portion **428**. The spring member **434** may bias the bracket member **402** and, hence the actuating member **302** in a locking position, as illustrated in FIG. 7. The high projections **420** may laterally restrict the bracket member **402** against the biasing of the spring member **434**. The spring member **434** is embodied as a helical spring. However, the spring member **434** may be any resilient component known in the art, for example, a rubber member, a spring washer, or the like.

(25) Referring to FIGS. 5, 8 and 9, a locking member **438** is attached to the third coupling portion **430** of the second lateral portion **406**. The locking member **438** is embodied as a pin including a head **440** and an elongate part **442**. The first lateral portion **404** includes an aperture **443** so that the elongate part **442** of the locking member **438** may be inserted into the circular aperture of the third coupling portion **430**. In an alternative embodiment, the locking member **438** may be integral with the second lateral portion **406**. The head **440** may retain the locking member **438** inside the circular aperture of the third coupling portion **430**. As illustrated in FIG. 8, in a locking position, the head **440** at least partly engages with the aperture **122** of the blade **102** when the blade **102** is in the retracted position. In the locking position, the locking member **438** may prevent the blade **102** from pivoting relative to the handle **104** since the bracket member **402** and, hence the second lateral portion **406** may not be allowed to pivot. The blade **102** is therefore retained in the retracted position. In the lower, middle and upper working positions of the blade **102**, the head **440** may engage the apertures **116**, **118**, and **120** (shown in FIG. 5), respectively, in order to retain the blade **102** in the respective working positions. In the locked position, the second lateral portion **406** of the bracket member **402** may be adjacent to the proximal end **105** of the blade **102**.

(26) FIG. 9 illustrates the locking member **438** in an unlocked position. As illustrated in FIG. 9, the actuating member **302** may be moved substantially along a sliding direction S. The inner boundary **308** and the central member **310** (illustrated in FIG. 7) of the actuating member **302** may press against the outer surface of the first lateral portion **404** of the bracket member **402**. Consequently,

the bracket member **402** also moves substantially along the sliding direction S against the biasing force of the spring member **434**. The second coupling portion **428** of the bracket member **402** slides over the connecting portion **432** (illustrated in FIG. 7) of the cover member **126**. The head **440** of the locking member **438** disengages from the aperture **122** of the blade **102** and the locking member **438** moves to the unlocked position. Thus, the blade **102** is free to pivot relative to the handle **104**. In the unlocked position, the first lateral portion **404** of the bracket member **402** may be adjacent to the proximal end **105** of the blade **102**.

(27) In an example, if a user wants to pivot the blade **102** from the retracted position (illustrated in FIG. 4) to the middle working position (illustrated in FIG. 1), the user may press the outer portion **305** of the actuating member **302**. The actuating member **302** may move along the sliding direction S and displaces the bracket member **402** in the direction S. Consequently, the locking member **438** may disengage from the aperture **122** of the blade **102**. The user may then pivot the blade **102** to the middle working position and release the actuating member **302**. The spring member **434** may bias the bracket member **402** such that the locking member **438** engages the aperture **118** of the blade **102**. The blade **102** may be therefore retained in the middle working position. In other examples, the user may want to pivot the blade **102** from the middle working position to any other position of the blade **102**. The procedure to be followed is substantially similar to the one described above.

(28) The blade **102** may therefore be conveniently adjusted by moving the actuating member **302** along the sliding direction S, pivoting the blade **102**, and then locking the blade **102** in the desired angular position. The inner boundary **308** and the central member **310** may increase an area of contact between the actuating member **302** and the bracket member **402**. Further, the outer portion **305** of the actuating member **302** may also provide an increased area to push the actuating member **302**. Thus, a user may have to exert lower force in order to move the bracket member **402** along the sliding direction S via the actuating member **302**. The grips **307** may also facilitate manual pressing of the actuating member **302**.

(29) As illustrated in FIG. 9, the locking member **438** may be completely disengaged from the blade **102** in the unlocked position. Thus, during pivoting of the blade **102** there may not be any contact between the blade **102** and the locking member **438**. This may substantially reduce wear of the various apertures **116**, **118**, **120** and **122** of the blade **102** and/or the locking member **438**. The user may also not feel reduced resistance and therefore be required to exert lower effort while pivoting the blade **102**. Further, the locking member **438** may not need any portion with a smaller width adjacent to the head **440** since the locking member **438** is completely disengaged from the blade **102** in the unlocked position. This may simplify a design and/or a manufacture of the locking member **438**.

(30) As illustrated in FIGS. 8 and 9, the locking member **438** may move substantially linearly along the sliding direction S. An angular movement of the locking member **438** may result in a smaller width of the head **440** of the locking member **438**. Therefore, the complete disengagement of the locking member **438** from the blade **102** and/or the substantially linear movement of the locking member **438** may allow a width of the head **440** that is close to a width of each of the apertures **116**, **118**, **120** and **122** of the blade **102**. Thus, a clearance between the head **440** of the locking member **438** and each of the apertures **116**, **118**, **120** and **122** may be minimized, resulting in a tighter engagement between the locking member **438** and the blade **102** in the locked position. Consequently, any undesired movement of the blade **102** in the locked condition may be substantially prevented.

(31) Further, in the locked position, the first and second lateral portions **404**, **406** of the bracket member **402** may be in contact with various other components, such as, the actuating member **302**, the horizontal planar portion **416**, the low projections **419**, the high projections **420**, the top projections **424**, the connecting portion **432**, the cover member **126** etc. This may prevent any undesired movement or pivoting of the bracket member **402**. Therefore, the locking member **438**,

that is coupled to the bracket member **402**, may be able to securely retain the blade **102**. Further, the proximal end **105** of the blade **102** may be at least partially enclosed by the channel **410** between the first and second lateral portions **404**, **406**. The second lateral portion **406** may also partly contact the proximal end **105** of the blade **102**. Thus, during an operation, forces acting on the blade **102** may be partly transmitted from the proximal end **105** to the various components in contact with the bracket member **402**. Consequently, an area of dissipation of the forces is increased. The bracket member **402** may enhance force transmission from the blade **102** to the handle **104**. Therefore, the forces may be dissipated without causing increased wear to the blade **102**, the locking member **438**, the bracket member **402**, and/or various other components of the tool **100**. Further, the ring member **202**, the screw **209** and the bushing **210** (illustrated in FIGS. 5 and 6) may result in easy pivoting of the blade **102** with minimum lateral movement. The bracket member **402** may also minimize any lateral movement of the blade **102** during pivoting. Forces acting on the blade **102** may also be partly transmitted to the ring member **202**, the screw **209** and the bushing **210**, thereby reducing wear.

(32) The handle **104** may be substantially a single component with the cover member **126**. The cover member **126** may be detachable to facilitate assembly and/or maintenance of the tool **100**. Single component configuration of the handle **104** may reduce any alignment errors during assembly. Manufacturing of the tool **100** may also be more cost efficient and simple. Further, the handle **104** may be compact and durable.

(33) In the drawings and specification, there have been disclosed preferred embodiments and examples of the invention and, although specific terms are employed, they are used in a generic and descriptive sense only and not for the purpose of limitation of the scope of the invention being set forth in the following claims.

PARTS LIST

(34) **100** Hand-Held Tool **102** Blade **104** Handle **105** Proximal End of the Blade **106** Cutting Edge of the Blade **107** Eyelet of the Blade **108** First Portion of the Handle **110** Second Portion of the Handle **112** End Support of the Handle **114** First Slot of the Handle **115** Second Slot of the Handle **116** Lower Working Position Aperture **118** Middle Working Position Aperture **120** Upper Working Position Aperture **122** Retracted Position Aperture **124** Non-Circular Aperture **126** Cover Member **127** Fastener **128** Hole of the Cover Member **129** Cover Coupler **130** Latch **202** Ring Member **204** Recessed Portion **206** Hole of the Ring Member **208** Hole of the Recessed Portion **209** Screw **210** Bushing **211** Head of the Screw **212** First Non-Circular Portion of the Screw **214** Second Non-Circular Portion of the Screw **216** First Circular Portion of the Screw **218** Second Circular Portion of the Screw **219** Stepped Portion of the Ring Member **220** First Circular Portion of the Bushing **221** Second Circular Portion of the Bushing **222** Hole of the Cover Member **224** Head of the Bushing **225** Blade Support Members **302** Actuating Member **304** Aperture of the First Portion **305** Outer Portion of the Actuating Member **306** Outer Surface of the First Portion **307** Grips of the Actuating Member **308** Inner Boundary of the Actuating Member **310** Central Member of the Actuating Member **402** Bracket Member **404** First Lateral Portion of the Bracket Member **406** Second Lateral Portion of the Bracket Member **408** Top Portion of the Bracket Member **410** Channel of the Bracket Member **412** Horizontal Bottom Edge **414** Inclined Bottom Edge **416** Horizontal Planar Portion **418** Inclined Planar Portion **419** Low Projections **420** High Projections **424** Top Projections **426** First Coupling Portion of the Second Lateral Portion **428** Second Coupling Portion of the Second Lateral Portion **430** Third Coupling Portion of the Second Lateral Portion **431** Projections of the Second Coupling Portion **432** Connecting Portion of the Cover Member **434** Spring Member **436** Inner Surface of the Cover Member **438** Locking Member **440** Head of the Locking Member **442** Elongate Part of the Locking Member **443** Aperture of the First Lateral Portion P Pivot Axis S Sliding Direction

Claims

1. A hand-held tool comprising: a handle; a blade having a proximal end pivotably coupled to the handle, the blade comprising a plurality of apertures; and a locking member configured to detachably engage with one of the plurality of the apertures to retain the blade at an angular position relative to the handle; wherein the locking member is provided on a bracket member coupled to the handle, wherein the bracket member is configured to at least partially enclose the proximal end of the blade; the bracket member is spring biased such that the locking member engages with one of the plurality of the apertures to retain the blade at an angular position relative to the handle; and the bracket member is movable such that the locking member disengages from one of the plurality of the apertures to allow the blade to pivot relative to the handle. I add., wherein the bracket member comprises two lateral portions, each lateral portion being connected to each other at one end, wherein the two lateral portions at least partially enclose the proximal end of the blade, and wherein a surface area of the two lateral portions of the bracket member facing the blade is greater than a surface area of the locking member facing the blade to dissipate forces from the blade via the bracket member. I addend..
2. The hand-held tool according to claim 1, wherein the bracket member comprises two lateral portions, each lateral portion being connected to each other at one end, wherein the two lateral portions at least partially enclose the proximal end of the blade..
3. The hand-held tool according to claim 1, wherein the locking member is integral with the bracket member.
4. The hand-held tool according to claim 2. I add. 1. I addend., wherein the locking member is a pin coupled to one of the lateral portions.
5. The hand-held tool according to claim 1, wherein the bracket member further comprises at least one first coupling portion, the at least one first coupling portion being configured to movably engage with a bracket support of the handle.
6. The hand-held tool according to claim 1, wherein the bracket member further comprises a coupling portion, the coupling portion being configured to engage with a connecting portion of the handle.
7. The hand-held tool according to claim 6 further comprises a spring member disposed between the bracket member and the handle, the spring member configured to bias the bracket member such that the locking member engages with one of the plurality of the apertures.
8. The hand-held tool according to claim 1 further comprises an actuating member configured to move the bracket member such that the locking member disengages from one of the plurality of the apertures.
9. The hand-held tool according to claim 8, wherein the actuating member at least partly projects from an outer surface of the handle.
10. The hand-held tool according to claim 8, wherein the actuating member comprises an inner boundary and a central member, the inner boundary and/or the central member being configured to contact the bracket member.
11. The hand-held tool according to claim 1, wherein the plurality of the apertures of the blade comprises: at least one working position aperture corresponding to a working position of the blade, the blade extending from the handle in the working position; and a retracted position aperture corresponding to a retracted position of the blade, the blade at least partially retracted within the handle in the retracted position.
12. The hand-held tool according to claim 11, wherein the handle comprises a slot configured to at least partially receive the blade in the retracted position.
13. The hand-held tool according to claim 1, wherein the blade further comprises a non-circular aperture, the non-circular aperture being configured to engage with a corresponding non-circular

portion of a screw.

14. The hand-held tool according to claim 13, wherein the screw engages with a bushing, the screw and the bushing being rotatable relative to the handle.

15. The hand-held tool according to claim 1, wherein the handle further comprises a detachable cover member.

16. The hand-held tool according to claim 1 wherein the hand-held tool is one of a handsaw or a knife.

17. A hand-held tool comprising: a handle; a blade having a proximal end pivotably coupled to the handle, the blade comprising a plurality of apertures; and a locking member configured to detachably engage with one of the plurality of the apertures to retain the blade at an angular position relative to the handle, the locking member having a first surface area facing the blade; wherein the locking member is provided inside a bracket member coupled to the handle, wherein the bracket member is configured to at least partially enclose the proximal end of the blade; the bracket member is spring biased such that the locking member engages with one of the plurality of the apertures to retain the blade at an angular position relative to the handle; and the bracket member is movable such that the locking member disengages from one of the plurality of the apertures to allow the blade to pivot relative to the handle, wherein the bracket member comprises two lateral portions, each lateral portion being connected to each other at one end, wherein the two lateral portions at least partially enclose the proximal end of the blade and have a second surface area facing the blade, and wherein the second surface area is larger than the first surface area.

.Iaddend.
